

Study on methanolytic depolymerization of PET with supercritical methanol for chemical recycling

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Abstract

Methanolytic depolymerization of polyethylene terephthalate (PET) was carried out in a stainless stirred autoclave at temperatures of 523–543 K, pressures of 8.5–14.0 MPa, and with a weight ratio of methanol to PET from 3 to 8. The solid products mainly composed of dimethyl terephthalate and small amounts of methyl-(2-hydroxyethyl) terephthalate, bis (hydroxyethyl) terephthalate, dimers and oligomers were analyzed by high performance liquid chromatography (HPLC). The liquid products composed of ethylene glycol and methanol were analyzed by gas chromatography (GC). It was found that both the yield of dimethyl terephthalate and the degree of PET depolymerization were seriously influenced by the temperature, weight ratio of methanol to PET, and reaction time, whilst the pressure has insignificant influence when it is above the critical point of methanol. The optimal depolymerization conditions are temperature of 533–543 K, pressure of 9.0–11.0 MPa, and the weight ratio (methanol to PET) from 6 to 8. The depolymerization of several PET wastes collected from the Chinese market was investigated under the optimal conditions. © 2001 Elsevier Science Ltd. All rights reserved.

Keywords: Polyethylene terephthalate (PET); Chemical recycling; Supercritical methanolysis; Depolymerization

1. Introduction

Polyethylene terephthalate (PET) is used extensively in the manufacture of high strength of fibers, photographic films and bottles for soft drink. With increasing PET consumption, its recycling has received wide attention for the preservation of resources and the protection of the environment. PET wastes can be melted and then reformed. However, the products have inferior properties and are limited in application. In contrast, chemical recycling can be realized by depolymerizing PET wastes into their original feedstock monomers, from which virgin resins with excellent performances are resynthesized [1–3].

Several processes for PET depolymerization have been put forward with different depolymerizing agents and operation conditions. Alcoholysis processes depolymerize PET to dimethyl terephthalate (DMT) with liquid or gaseous methanol [4–6], glycolysis depolymerizes PET to bis(hydroxyethyl) terephthalate (BHET) with ethylene glycol

or other glycol [7–9], and hydrolysis [10–13] to terephthalate acid (TPA) under the promotion of acidic or basic conditions. All of these processes have both advantages and disadvantages. Methanolysis under normal conditions can be carried out at relative mild temperature and pressure. However, the reaction rate is very slow, and some divalent metal catalysts such as zinc, lead, and manganese acetates, are required to enhance depolymerization rate. The undesired diethylene glycol, the dimer of ethylene glycol, is formed in the glycolysis. There are thus some problems in separation and purification of the product [14]. Hydrolysis under acidic or basic conditions may cause corrosion and pollution problems [15], whilst both terephthalic acid formed in the reaction and the acidic catalysts favour the formation of diethylene glycol. Some new experimental methods have been introduced for the decomposition of PET. As an example, microwave irradiation was used as energy source in solvolysis of PET [16], and a high-pressure calorimeter has been used for measure the decomposition degree [17].

In recent years, the supercritical fluids, supercritical water ($T_c = 674.3$ K, $P_c = 22.0$ MPa) [18] and supercritical methanol ($T_c = 512.3$ K, $P_c = 8.09$ MPa) [19], have been introduced to the depolymerization of PET.

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PET hydrolysis with supercritical water has very high reaction rate, but this process is not easy to operate in practice due to the severe reaction conditions (above 670 K, 30 MPa). In addition, the hydrolysis leads to low yield of ethylene glycol (about 20%). Compared with supercritical hydrolysis, supercritical methanolysis can be operated at relatively mild conditions. Sako et al. [19] proposed the methanolytic depolymerization of PET to DMT and ethylene glycol under the supercritical state of methanol, and investigated the influence of the reaction pressure and time on the yields of decomposition products at a relative higher reaction temperature (573 K). Subsequently, they determined the rate constant of decomposition of PET with methanol from 453 to 623 K [20]. However little effort has been made to investigate the effect of other degradation parameters on the depolymerization of PET.

In this study, a series of experiments were conducted to further study the methanolytic depolymerization of PET, particularly to determine the effects of the temperature, pressure, reaction time, and weight ratio of methanol to PET on the depolymerization degree and DMT yield. A new analysis method with high performance liquid chromatography was proposed to analyze the content of components in the solid products [21,22]. The optimal reaction conditions were obtained. Under the optimal conditions, the depolymerizations of several kinds of PET wastes collected from Chinese market were tested.

2. Experimental

2.1. Materials

The pure PET resins were commercial chips of fiber grade supplied by Liaoyang Petrol-Chemical Fiber Company (Liaoyang City, PR China). The intrinsic viscosity (IV) of PET [measured in a 60:40 (w/w) phenol/1,1,2,2-tetrachloroethane solution at 298 K] is 0.661 dl/g corresponding to the viscosity molecular weight (M_v) of 19,700, which is calculated from the equation of $[M_v] = 3.61 \times 10^4 IV^{1.46}$ [23]. Analytical grade methanol and other reagents were bought from Tianjin Chemical Reagent Factory (Tianjin City, PR China), and the methanol was used after further purification by distillation. Standard samples of BHET (99.5%) and DMT (>99.5%) were obtained from Aldrich, USA, and standard sample of MHET was synthesized according to Yamanis's method [24].

Several kinds of PET wastes were collected from the Chinese market for the depolymerization test. All were firstly washed by hot deionized water (353 K), broken into chips with dimensions of 3–5 mm, and then dried at 323 K to get rid of water and volatile compounds before being loaded into the reactor.

2.2. Methanolysis

A 150 ml stainless batch-type autoclave equipped with an electromagnetic stirrer, thermometer well, inert gas inlet and outlet valves was used for the methanolysis. A high-power heating collar controlled by a programming temperature controller was used to heat the reactor rapidly to the reaction temperature. The deviation of the temperature in the reactor was within about ± 1.0 K. The temperature of reaction system was measured by a K-type thermocouple.

For the tests, PET and methanol in a certain ratio were charged into the reactor at room temperature. Before heating, a small flux of inert gas, such as N_2 or Ar, was introduced to the reactor for replacing air to avoid the possible oxidation of PET and depolymerization products. The reactor was then heated as fast as possible to the temperature specified for conducting the methanolysis reaction. It took about 10 min to heat up the reaction system to the critical temperature of methanol, and 1–2 min from the critical point to the given reaction conditions. After the required time of methanolysis at designed reaction conditions above the critical point of methanol, the vessel was removed from the heating collar, and quenched in a water-ice bath. The temperature of the reaction mixture dropped to under 333 K in 1–2 min of quenching. After the reactor vessel was cooled to about 298 K, it was opened and the products in the reactor were then taken out. A certain amount of cold water (273–283 K) was added to the mixed product to dissolve the small quantity of ethylene glycol absorbed on the surface of solid products and extract out some components dissolved in methanol such as DMT and small amount of methyl-(2-hydroxyethyl) terephthalate (MHET). The mixture was subsequently separated into solid and liquid phases by filtration. The contents of methanol, ethylene glycol and water in the filtrate were then analyzed with GC. The solid product mainly composed of DMT, MHET, BHET, dimers and oligomers was then dried at 323 K to a constant weight, and ground to a fine powder for HPLC analysis.

Depolymerization experiments for waste PET chips were performed to recover the DMT and ethylene glycol. As in the above-mentioned process, the mixture of methanol and waste PET chips with weight ratio of 6 were added to the reactor, and depolymerized at 543 K, 11.0 MPa for 40 min. After that, the mixture products were treated in a similar manner. The dried powder of solid product was partly used for HPLC analysis, the rest was purified by recrystallization as follows: 5 g powder was added to a certain amount of methanol to solubilize DMT at 313 K, then the solution was filtered and the filtrate distilled to dryness at 323 K to give almost pure DMT 4.60 g in 92% recovery. The purified solid product was kept for characterization.

2.3. Analysis and characterization

The solid samples were analyzed by RP-HPLC using a Shimadzu LC-3A high performance chromatograph (Japan) equipped with a reverse-phase Zorbax-C₈ (4.6×250 mm) column and a SPD-1 ultraviolet detector set at 254 nm. The RP-HPLC procedure was as follows: 20–40 mg/l sample solution for analysis was prepared by dissolving a certain amount of the dried solid product in a 70:30 (V/V) methanol/water solution, which was also used as mobile phase with the flow rate of 1 ml/min. The temperature of the column was room temperature (298 K). The components of DMT, MHET and BHET were quantified by external standard method, while dimers and oligomers were quantified by inner unitary method (The inner unitary method is commonly called normal chromatographic quantitative method. For the components with near structure, it can be assumed that the chromatographic correction factor is 1, and the amounts of component are quantified with this assumption.)

The concentrations of the components in the liquid products were analyzed by gas chromatography (GC). The liquid sample was injected into a GC-920 (Shanghai, PR China) gas chromatograph and separated by a 2-m stainless column packed with PEG-20M. The components were carried by hydrogen at the flow-rate of 30 ml/min and detected by an FID. The temperatures of the column, detector and injector port were 433, 473 and 473 K, respectively.

The measurement of elemental composition of carbon and hydrogen in the purified samples was performed by a Perkin-Elmer 2400 elemental analyzer, USA. IR spectra of purified DMT was measured on a 25PC FT-IR spectrograph (Bio-Rad, USA) by KBr method.

3. Results and discussion

3.1. Calculation of reaction extent

Since the products obtained from supercritical methanolysis were mainly DMT and small amounts of byproducts, MHET, BHET, dimers and oligomers, it can be assumed that the methanolysis in the supercritical methanol proceeds with the same reaction mechanism as that in the liquid methanol.

As shown in Fig. 1 each ester linkage uses a methanol molecule and creates the end groups of a methyl ester and ethylene glycol. Secondary decomposition of the product monomers was neglectable because of the low depolymerization temperature of 523–553 K. At same time, the byproducts MHET, BHET, dimers and oligomers are formed according to the above-mentioned reaction scheme, in which the ester linkages of ethylene glycol in these byproducts are still not broken, thus it is impossible to accurately quantify the amount of

depolymerized ester linkages in PET chains. Therefore, the degree of PET depolymerization is here defined as the ratio of the quantity (moles) of consumed methanol to the moles of the ester linkages in the PET [Formula (1)]. The mole of methanol is defined as Formula (2). The yield of DMT is represented as the percentage of the DMT amount in the solid products calculated from HPLC data against the theoretical DMT amount if the feed PET was completely depolymerized [Formula (3)].

Degree of PET depolymerization (%)

$$= \frac{\text{Moles of consumed methanol} \times 100}{\text{Moles of ester linkages in PET}} \quad (1)$$

Moles of methanol =

$$\frac{\text{The content of methanol in filtrate} \times (\text{total weight of product} + \text{weight of added water} - \text{weight of dried solid})}{\text{Molecular weight of methanol}} \quad (2)$$

Yield of DMT (%) =

$$\frac{\text{Weight of actual output DMT monomer} \times 100}{\text{Theoretic output}} \quad (3)$$

From a blank test, it was found that there were about 5.0% ester linkages broken during the heating interval before reaching the required temperature, while the content of DMT monomer in the solid product was found to be less than 0.2 wt.%. This indicates that initial degree of PET depolymerization for the reaction under supercritical conditions is about 5.0%, rather than zero. As a matter in fact, this initial depolymerization effect was considered in the later discussion of the reaction performance under different conditions.

3.2. Analysis of solid products

With the HPLC method the calibration curves of components (DMT, MHET and BHET) in the solid product are all linear within the range of 1.0~45 µg/mL ($n=10$), correlation coefficient $r=0.9998\sim1.0000$, and the relative standard deviation (R.S.D.) of components are less than 2.8%. The determination limits of composites was 4.0×10^{-4} µg. Fig. 2(a)–(c) shows HPLC traces of solid products with different depolymerization times (10, 20 and 60 min) at 533 K, 11.0 MPa, the weight ratio of methanol to PET in feed is 6. It is clear that the composites of DMT, MHET, dimer of BHET, dimer of DMT, oligomers and small amount of BHET in the solid sample could completely be separated and detected accurately by using this method. Regarding the reaction condition at 533 K, 11.0 MPa, 60 min, the amounts BHET, dimers and oligomers were too small to be detected, and the

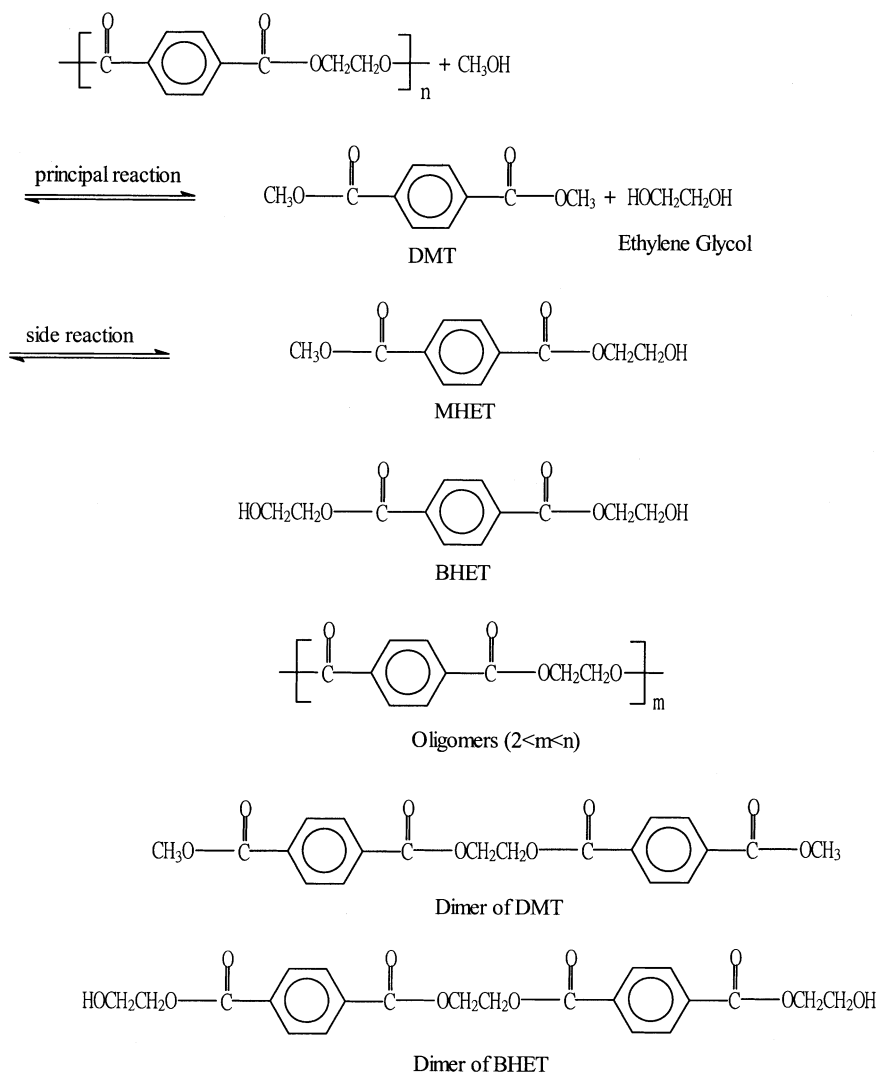


Fig. 1. Overall reactions of PET depolymerization in supercritical methanol.

main component was DMT monomer. Only a little MHET could be detected. At 533 K, 11.0 MPa, 10 min, the amount of MHET increased, and the peaks of BHET, dimers and oligomers were observed corresponding to the shortening of the depolymerization time.

3.3. Effects of reaction parameters

The effect of weight ratio methanol to PET on depolymerization at 533 K, 11.0 MPa for the reaction time of 20 min is summarized in Fig. 3. The result indicates that the degree of PET depolymerization and the yield of DMT increase with the weight ratio of methanol to PET. It is possible that PET can be well dispersed in the supercritical methanol when the ratio of methanol to PET is high. It should be noticed that with the increase of the methanol to PET ratio, the depolymerization degree of PET and the DMT yield increase sharply when the ratio of methanol to PET is less than 6, and then increase slowly when the ratio is above 6. The results indicate

that when the reaction time is about 20 min, the methanol to PET ratio should be higher than 6 in order to gain the optimal depolymerization results. It implies that under constant temperature, pressure and reaction time, there exists an optimal weight ratio of methanol to PET on the basis of whole effectiveness.

To study the influence of the reaction temperature and reaction time on the depolymerization, the experiments were carried out at 11.0 MPa, 523, 533 and 543 K with the reaction time of 10, 15, 20, 25, 30, 40, 60 min and the methanol to PET ratio (weight) of 6. Fig. 4 shows the plots of the degree of PET depolymerization versus reaction time at different depolymerization temperatures. When the reaction time was less than 40 min, PET was not completely depolymerized. The depolymerization degree was notably enhanced by the increasing temperature. After being depolymerized for 40 min, nearly all PET had been depolymerized. Fig. 5 shows the plots of the yield of DMT versus reaction time. The effects of temperature and reaction time on the DMT yield were

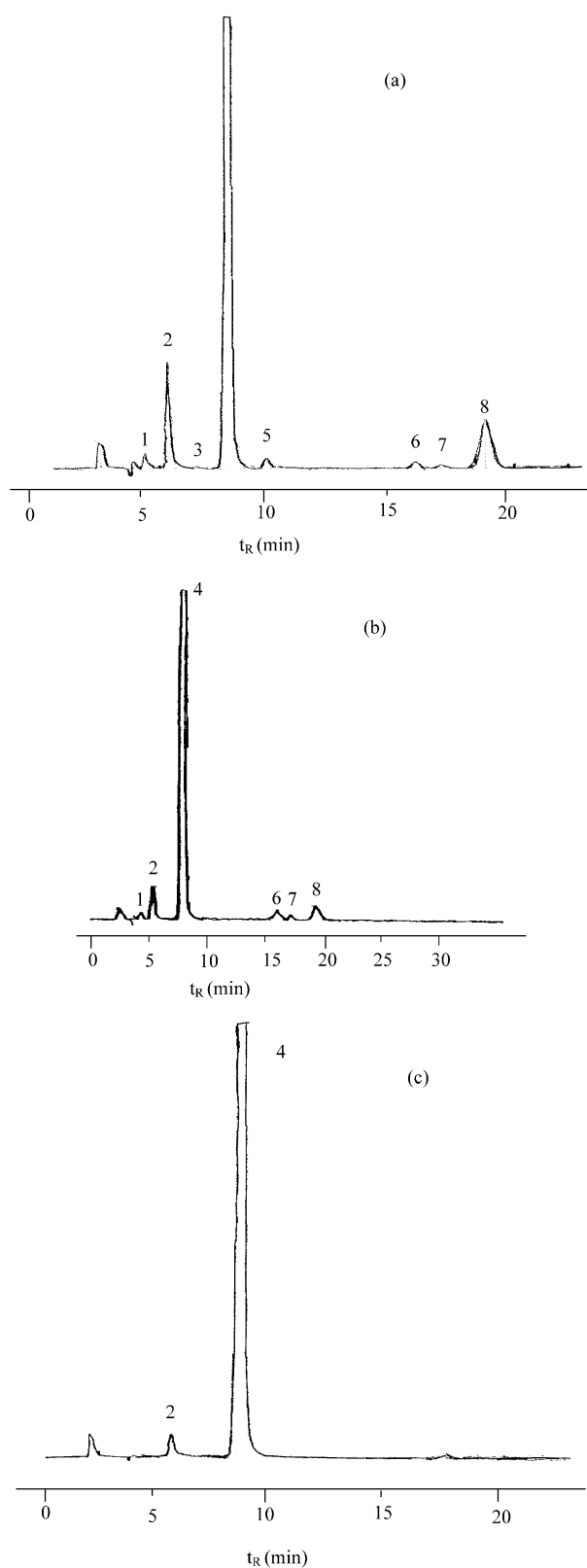


Fig. 2. HPLC spectra of solid products after different reaction time at 533 K, 11.0 MPa and the weight ratio (methanol to PET) of 6 (a) 10 min; (b) 20 min; (c) 60 min. (1) BHET; (2) MHET; (3) BHET dimer; (4) DMT; (5) DMT dimer; (6–8) oligomers.

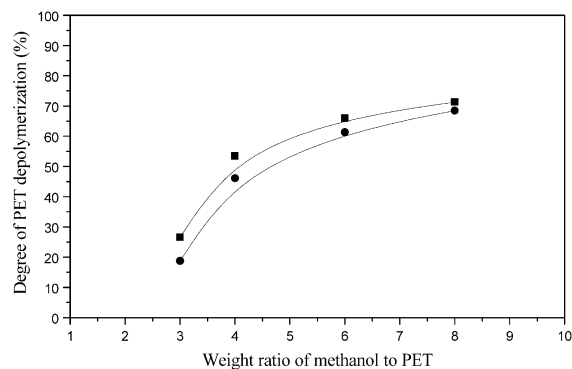


Fig. 3. Effect of the weight ratio of methanol to PET on the yield of DMT and the degree of PET depolymerization at 533 K, 11.0 MPa for the reaction interval of 20 min. (■) the degree of PET depolymerization, (●) the yield of DMT monomer.

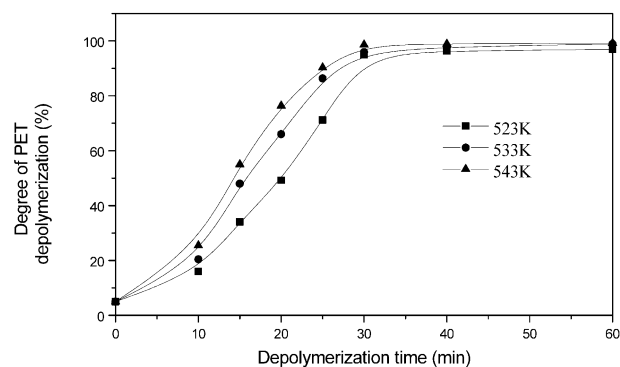


Fig. 4. Effect of reaction time on the depolymerization degree of PET at 11.0 MPa, the weight ratio of methanol to PET of 6.

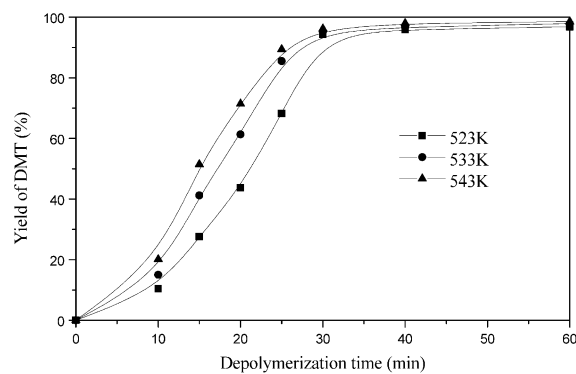


Fig. 5. Effect of reaction time on the yield of DMT monomer at 11.0 MPa, the weight ratio of methanol to PET of 6.

same as that of depolymerization degree. The yield of DMT monomer increases with increasing reaction time and temperature before the reaction time of 40 min. Beyond that, there are no obvious increases in either degree of PET depolymerization or the yield of DMT monomer, and PET approaches complete depolymerization.

During the methanolysis of PET, several byproducts such as MHET, dimers (including two repeated units), oligomers (including three or more repeated units) and minute quantities of BHET are formed [25]. Fig. 6 illustrates the change of the yields of those byproducts versus reaction time at 533 K, 543 K and 11.0 MPa with the methanol to PET ratio of 6. Due to the conversion of byproducts to the monomers, the yields of those byproducts rapidly decrease with the increase of reaction time as well as the increase of temperature. These results indicated that PET could be completely depolymerized within 40 min, at a temperature of 523–543 K.

To investigate the effect of pressure on the depolymerisation behavior of PET around the critical point of methanol, a series of experiments with different reaction pressure varied from 7.0 to 12.0 MPa were performed at 533 K for 20 min, with weight ratio of methanol to PET of 6. The pressure of reaction system was controlled by adjusting the added amounts of methanol and PET. Table 1 displays the effect of reaction pressure on the

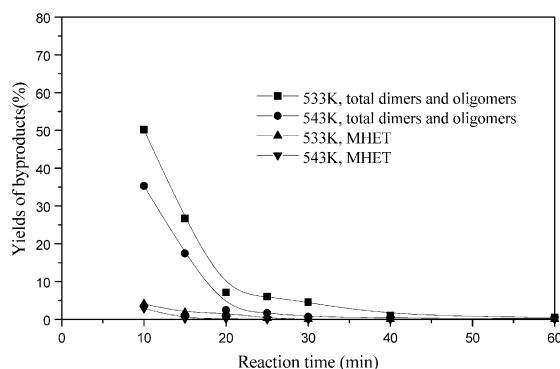


Fig. 6. Effect of the reaction time and temperature on the yields of byproducts at 11.0 MPa, the weight ratio of methanol to PET of 6.

Table 1
Effect of reaction pressure on the depolymerization at 533 K for 20 min, the weight ratio (methanol /PET) of 6:1

Pressure (MPa)	7.0	9.0	10.0	11.0	12.0
Depolymerization degree of PET (%)	20.50	63.00	65.50	66.10	64.45
The yield of DMT (%)	6.41	50.01	59.10	61.33	61.30

Table 2
Results of methanolysis for different PET wastes at 543 K, 11.0 MPa for 40 min with the weight ratio (methanol to PET) of 6:1

PET wastes	Degree of PET depolymerization (%)	The purity of DMT in unpurified product (%)	The purity of DMT in purified product (%)
550-ml pure water bottle (transparent) (Wahaha Co. Ltd., Hangzhou City)	98.71	96.53	98.90
550-ml mineral water bottle (transparent) (Robust Co. Ltd., Zhongshan City)	97.97	96.72	98.85
1.25-l Spirit bottle (green) (Taiyuan Coca-Cola Co.Ltd.,Taiyuan City)	96.54	95.32	98.01
1.25-l Coke bottle (transparent) (Taiyuan Coca-Cola Co.Ltd.,Taiyuan City)	98.23	96.47	99.02
Waste filum (Taiyuan Fiber Factory, Taiyuan City)	96.78	95.19	98.13

degree of PET depolymerization and the yield of DMT monomer at 533 K with the methanol to PET ratio of 6 for the reaction time of 20 min. It shows that the pressure has little influence on the degree of depolymerization and the yield of products when the reaction pressure surpassed the critical pressure of methanol, while the pressure has strong influence around the critical point. It is because that the solvent density is high under the supercritical state, nearly that of normal liquid, and this gives the solvent a high capacity for solute [26]. For this reaction system, methanol has higher capacity for dissolving PET under the supercritical state of methanol than that of normal liquid methanol. The active ester linkages in PET could be well dispersed in the methanol and it facilitated the depolymerization. On the other hand, the reaction in the supercritical state has a big negative partial molar volume change near the critical pressure, which can increase the reaction rate constant significantly. To further increase the pressure, the partial molar volume change tends to zero, and the reaction pressure has small influence on the reaction [27]. This result indicated that the pressure is not a key factor of degradation of PET under the supercritical state of methanol. However, the reaction pressure has strong effect on the depolymerization near the critical point, which agrees very well with the result reported by Sako and his co-workers [19].

3.4. Depolymerization of PET wastes

It is shown from the above-mentioned depolymerization results of pure PET that the optimal depolymerization conditions are a reaction time of 40–60 min, a temperature of 533–543 K, pressure of 9.0–11.0 MPa, and weight ratio (methanol to PET) from 6 to 8. Under the optimal depolymerization conditions, we have also investigated the depolymerization of waste film and used bottle flakes of several main brand beverage bottles in Chinese market. The depolymerization of those PET wastes was carried out for 40 min at the conditions of 543 K, 11.0 MPa, and methanol to PET ratio of 6. The results of depolymerization are summarized in Table 2. It indicates that the various PET wastes can be well depolymerized under the optimal reaction condition. the

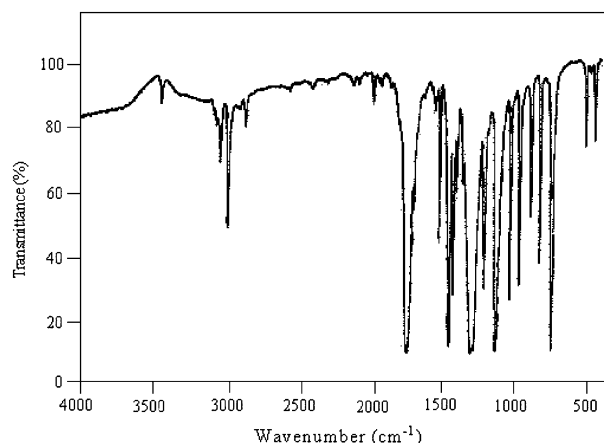


Fig. 7. IR spectrum of purified sample.

Table 3
Elemental analysis of purified sample (wt. %)

	C	H	O ^a
DMT (calc.)	61.86	5.15	32.90
Sample (obs.)	61.79	5.17	33.04

^a By difference.

color and the source had no distinct influence on the depolymerization.

After depolymerization, to obtain the purified solid sample of DMT monomer and remove oligomers and other impurities from the solid products of depolymerization, the solid product was purified according to the method mentioned in the experimental part. By purification with methanol, the DMT purity in the product can reach 98%. The melting point of the purified sample is 414–415 K, which is same as that of pure DMT. Fig. 7 gives the IR spectrum of the purified sample from the depolymerization of waste from 'Robust' bottle. The IR spectrum indicates a pure DMT structure. The absorptions at 3000–2800, 1750, and 1100 cm^{-1} are assigned to C–H, C=O and C–O stretching vibrations, respectively. This result agrees with the standard spectroscopy of DMT [28].

The elemental analysis of the purified sample is listed in Table 3. The contents of C, H and O (by difference) coincide well with the calculated values of pure DMT monomer. Other purified samples have the similar results with that of 'Robust' bottle.

4. Summary and conclusions

PET polyester can be rapidly and completely decomposed to its monomer DMT under the supercritical state of methanol. The degree of depolymerization and selectivity for dimethyl terephthalate increased with an increase of the ratio of methanol to PET, reaction temperature, and

reaction time, while pressure had little influence on either of them when the pressure of the reaction exceeds the critical pressure of methanol. According to results of our experiments, the optimal reaction conditions are reaction time of 40–60 min, temperature of 533–543 K, pressure of 9.0–11.0 MPa and weight ratio (methanol to PET) of 6–8. Under the optimal reaction conditions PET wastes can readily be depolymerized to monomers. The technique of reversed-phase high performance chromatography and gas chromatography were used to analyze the solid and liquid products respectively, and the different components in the product can be separated well and analyzed accurately for the quantification.

Acknowledgements

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